

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2024–2025
FULL MARKS: 75

CSE 4635: Web Architecture

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. **Scenario: CampusBites** A food delivery startup wants to build a web-based food ordering platform with the following features:

- Customers can register, log in, browse restaurants, search by cuisine, place orders, and view order history
- Restaurant owners can manage their menus, view incoming orders, and update order status.
- The admin can add or remove restaurants and monitor platform-wide analytics (e.g., daily orders, popular cuisines).
- Order confirmation and delivery status updates should be sent asynchronously.
- The system must handle peak-hour traffic (e.g., lunch time) and scale as more restaurants join the platform.

a) What is web architecture? Explain with suitable examples the difference between a 2-tier and a 3-tier web architecture.

2 + 3
(CO1)
(PO1)

b) Based on the CampusBites scenario described above, answer the following.

i. Describe the complete HTTP request-response lifecycle when a customer places an order through the application. Your answer should include:

5
(CO1)
(PO1)

- The HTTP method used
- At least two relevant request headers
- Expected HTTP status codes for success, validation failure, and server error.

ii. CampusBites needs to maintain user sessions after login. Compare **URL Rewriting** with **HttpSession** for managing the customers login state. Which approach would you recommend for CampusBites and why?

5
(CO2)
(PO1)

c) Write a Java Servlet **OrderServlet** that handles order placement for CampusBites. The servlet should:

10
(CO2)
(PO2)

- Accept POST requests with parameters: **customerId**, **restaurantId**, and **items** (comma-separated item IDs)
- Validate credentials against hardcoded values (e.g., **customerId = "cust01"**)
- Check if the customer is logged in via session validation; if not, redirect to **login.html**
- If logged in, process the order and forward to **orderConfirmation.jsp**.

Include the appropriate servlet annotation or **web.xml** mapping.

2. a) Write short notes on the following concepts. For each, include an example. 3 × 3
(CO1)
(PO1)
- i MVC Pattern (Model-View-Controller).
 - ii Inversion of Control (IoC) Container.
 - iii Tight Coupling vs Loose Coupling.
- b) Given the following considerations for CampusBites (scenario given in Question 1): 6
(CO4)
(PO2)
- Current scope (single campus, small development team).
 - Future expansion to multiple university campuses.
 - The need for handling peak-hour traffic during lunch time.
- Would you recommend a Monolithic or Microservice architecture at this stage? Justify your answer.
- c) For the CampusBites application, draw an architectural diagram that shows the placement and interaction of the major components within your chosen architectural style. 8
(CO3)
(PO3)
3. a) Design RESTful API endpoints for the following features of CampusBites. For each endpoint, specify the HTTP method, URI path, request body (if applicable), and expected response status code. 2 × 4
(CO2)
(PO2)
- i Browse restaurants by cuisine type.
 - ii Place a new order.
 - iii Get all orders for a specific customer.
 - iv Update the status of an order (e.g., preparing delivered)
- b) Using Spring Boot, write code for the following: 5 + 10
(CO3)
(PO3)
- i. A Restaurant model class with the following fields: *id* (Long), *name* (String), *cuisine* (String), *rating* (double). Use appropriate annotations.
 - ii. A **RestaurantController** class with the following endpoints:
 - GET /api/restaurants returns all restaurants.
 - GET /api/restaurants/id returns a specific restaurant by ID.
 - POST /api/restaurants creates a new restaurant.

Use proper Spring Boot annotations (@RestController, @RequestMapping, @GetMapping, @PostMapping, @PathVariable, @RequestBody). Assume a **RestaurantService** class is available and injected via constructor injection.
- c) Briefly explain how the CAP (Consistency, Availability, Partition Tolerance) theorem applies in the CampusBites application when multiple users attempt to book the same seat at a restaurant simultaneously. Which two of the three CAP properties would you prioritize? Justify your choice. 4
(CO1)
(PO1)